

Power BI

(Power BI is a king of dashboard)
(Power BI is a tool of microsoft)
[Business Intelligence]

* Dashboard complete process :-

- 1) Data loading
- 2) Transformation (in power query or power editor)
(cleaning, editing)
- 3) Data modeling (relationship)
- 4) Dax formulas
- 5) Visualization

* use of dashboard :- analysis of data, data visualization, and report sharing.

* what is Power BI :-

Power BI is a business intelligence tool developed by microsoft that helps convert raw data into meaningful insight through interactive dashboard, visualization, and report.

* Components of Power BI :-

- Power BI Desktop :

The main tool used to create reports and dashboards.

- Power BI service (web) :

cloud-based platform to publish, share and collaborate.

- Power BI mobile app :

To view dashboard on mobile devices.

- Power BI Gateway :

Connects on-premises data with cloud services. (for automatically updated)

- Power BI report server :

on-premise solution for hosting reports.

hierarchy - where things are ranked by importance

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* Who needs to learn BI :-

- Business Analysts:
To visualize business data and KPIs
- MIS Executives: (management information system)
for automated, dynamic reporting.
- Data analysts:
To convert raw data into interactive dashboards.
- Team leaders / Managers:
To monitor team performance via dashboards.
- Entrepreneurs & freelancers:
To create visual business performance reports.
- Students & Job seekers:
To build a strong resume for data-related roles.

* Hierarchy in data field :-

- MIS executive:
manage day-to-day reporting and excel dashboard.
- Data analyst:
performs data cleaning, visualization and basic forecasting.
- Business analyst:
focuses on business strategy using data insights.

- Data mining expert:
extracts patterns using algorithm and large data sets.
- Data scientist:
Builds predictive models and works on machine learning.

* Workflow of power BI

Step 1 : Loading Data

Step 2 : Transformation (editing, cleaning data)

Step 3 : Data modeling (relationship)

Step 4 : Dax formula (measures)

Step 5 : Visualization (Dashboard)

* Loading queries

Load data → importing data in power BI
(This is called import mode query)

Source data → we do not import data
(like sql data) in power BI

Source data त्याच्या ठिकाणावरून हलवला जाही जात तो आहे निघेच ठेवला जातो. data transformation पण आहे निघेच होत power BI मध्ये पावत dashboard वापरला use source data used to live visualization on dashboard and it is interactive (this is called direct query)

power BI keeps updating connectors.

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check - 'Get Data' → 'More option'

• Data connectors :-

(no plus
connecti-
vely)

- Power BI offers a wide variety of data connectors to import data from multiple source.
- connectors act as a bridge between your data source and power BI.
- Eg., excel, text/csv file, SQL server database or so on.

• Data Loading :-

- In power BI, Data open in power query
- Load : only load data in power BI
- Transform data : process on data like cleaning
- one time, we can one or more data load

* Practical of loading data in Power BI

open Power BI desktop

open blank report

click on 'get' data

select data

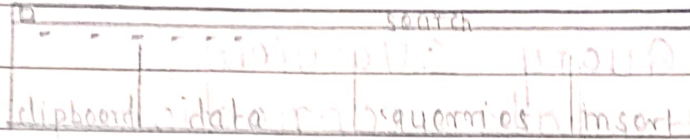
then click on 'load' option

• Interface overview :-

- Extension of power BI is .pbix
- on blank report file, home, insert, modeling, view, optimize this is called Tab

Power query ^{editor} data - after transform data, data go in power query editor

- clickboard, data, queries, insert, calculation sensativity & share, copilot. this is called group. In inner of group their are a features.



} called Ribbon

- Visualization and data are fixed then It is called visualization pane or data pane.

- filters helps seen data in various type.

-  — Desktop mode

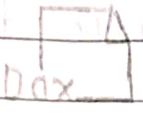
-  — mobile mode

-  (Report view)

Dashboard is made on report view

-  (Table view)
seen's all data we loaded

-  (model view)
relationship between data.

-  (Dax View)
write formula or queries.

- Transformation Query -
After loading, the step of transfor-

person, which work on source (direct)
query it also called work on
Native query (work on live data)

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 Use of Import query

we work on power BI for transformation
their is called M query (manual query)

• Power Query Overview:

- It is a data connection and transformation tool used in power BI Desktop. It follows ETL (extract, transform, Load) principles.
- power query editor in power BI is used for cleaning, transforming and shaping data before it's loaded into the model.

Types of

Transformations in
power
Query

• Column level

[We also
conceded
step
which
applied
on data]

- 1) rename columns - change column headers.
(ctrl + F2)
- 2) remove columns - delete unwanted columns.
(right click)
- 3) reorder columns - adjust column order.
(Drag)
- 4) split columns - split by delimiter, number of character, etc.
(data tab)
- 5) merge columns - combine multiple columns into one.
(data tab)
- 6) change data type - text, number, date, etc.
- 7) replace values - replace specific text or numbers.

[Delimiter
is a
delimiter]

Transpose - Vertical data ^{horizontal} data.

Flip - 1 to 100 $\rightarrow$ 100 to 1

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- Row - Level - Transformations :-

- 1] Remove rows - Top rows, bottom rows, duplicates, blank rows.
- 2] keep rows - Top rows, range of rows, duplicates, etc.
- 3] filter rows - use conditions to filter out data.
- 4] Group By - aggregate rows (sum, average, count).
- 5] sort rows - ascending or descending order.

- Table - Level Transformations :-

- 1] Transpose Table - flip rows and columns.
- 2] unpivot columns - convert column headers into row values.
- 3] pivot columns - Turn row values into column headers.
- 4] Remove errors - Delete error rows or cells.
- 5] Detect data type - auto - detect best type for each column.

- Add column Transformations :-

- 1] custom column - use M language formulas to create new columns.
- 2] Index column - add serial numbers.
- 3] conditional column - add column based on if-else logic.
- 4] Duplicate column - copy an existing column.

column from example
(give some logic then power BI
detect it easily)

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No option

70 power BI

5) Invoke custom function - apply a function
on a data

• Text Transformations -

- 1) Trim / clean - remove extra spaces or non-printing characters.
- 2) uppercase / lowercase / capitalize - modify text casing.
- 3) Extract Text - from start, end, or a specific position.
- 4) Replace Text - change parts of text in a column.

• Number Transformations -

- 1) Standardize - round, ceiling, floor, absolute, etc.
- 2) Statistics - minimum, maximum, average, count (Group By Feature)
- 3) Add / subtract / multiple / divide - perform custom columns - math on columns.

• Date & Time transformations -

- 1) Date only / time only / year / month / day - extract parts of date.
- 2) add / subtract days - shift dates.
- 3) ago calculation - calculate difference from today.
- 4) Date difference - days between two dates.

• advanced transformations -

- 1) parameters - create dynamic filters or control values.

(Mashup Language)
[M Language - formally known as Power Query M formula language, is the primary language used in data transformation and preparation in Power BI]

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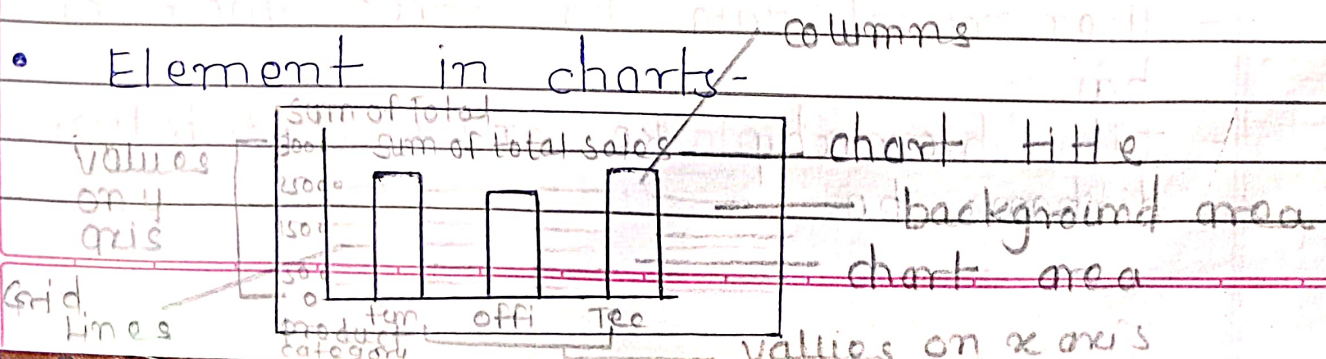
- 2) merge queries - like SQL JOIN (inner, left, right, full)
- 3) append queries - combine data from multiple tables
- 4) use first row as headers - promote header rows
- 5) reference vs duplicate query - reuse logic (view feature) smartly

• Visualization Overview -

* clustered column, bar chart -

- clustered column chart -
 - multiple categories ke data ko vertical bars ke form mein compare karta hai
 - Jab apko alag-alag categories ka comparison dikhana ho
 - category aur numerical values
eg. sales

- clustered Bar chart -
 - column chart ka horizontal version.
 - Jab categories long ho ya names bade ho
 - category aur numerical value.



* stacked column and bar -



- stacked column chart -
- ek hi bar mein sub-categories ko stack karke dikhata hai
- Jab total ke andar har part ka contribution dikhata ho
- category, sub-category, numeric



- stacked bar chart -
- bar chart jisme sub-categories stack hoti hain
- same as stacked column, par horizontally
- category, sub-category, numeric

* 100% stacked - column and bar -

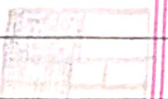


- column -
- Har column ko 100% ke base par stack karta hai
- Jab proportion (percentage) dikhani ho
- category, sub-category, percentage values



- bar -
- bar ko 100% base mein divide karta hai
- Jab percentage compare karna ho
- category, sub-category, numeric

* Tree Map - (Boxes format)



- Hierarchical data ko boxes mein dikhata hai
- Jab hierarchy ke andar contribution dikhana ho

Stack - category inner details.
clustered - positioned close together
of pillars, columns.

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- category / sub-category, values.

* Line, area and stacked chart -

Area

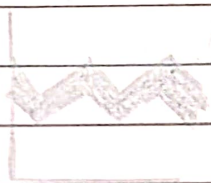
• Line chart -

- Line chart jaisa hi bas area fill hota hai
- Trend + volume emphasis dikhani ho
- Date / time, numerical value.

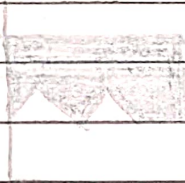
• Line chart -

- Time series data ke liye use hota hai
- Jab trend dikhana ho across time
- Date / time, numerical value.

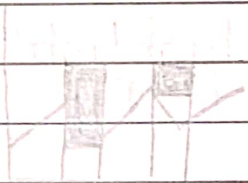
Stacked
area
chart



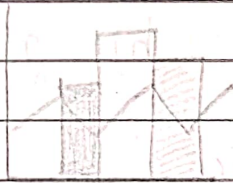
100% stacked
area chart



Line and
stacked
column
chart



line and
clustered
column chart



* Pie and donut charts -

Pie chart -

- - Total ka part dikhata hai in circular format.
- Jab small number of categories ho
- category, value.



• donut chart -

- pie chart with a hole in center.
- same as pie but with modern look.
- category, value.



Pivot-changing rows into columns
unpivot - opposite of pivot -
turning columns back into rows

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* Table & Matrix -

• Table -

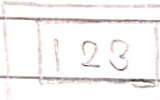
- 
- raw data ko tabular format mein dikhata hai
 - detailed view chahiye ho bina visuals ke
 - multiple columns (category & values)

• matrix -

- 
- pivot - table jaise rows & columns mein dikhata hai
 - Jab summarised view with groupings chahiye
 - row, column fields, value field

* card & multiple card (row) -

• card -

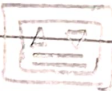
- 
- single value dikhata hai
 - KPI ya ek hi number highlight karna ho
(KPI - keep performance indicators)
 - single metric

• multi-row card -

- 
- multiple single values dikhata hai
 - Jab ek saath kai KPI dikhana ho
 - multiple metrics

* KPI and gauge -

• KPI -

- 
- Target vs actual with trend line
 - Jab goal tracking dikhana ho
 - actual value, target, date

• Gauge -

- Value ko dial format mein dikata hai
- Jab performance range dikhami ho
- actual, min, max.



* charts / graph:-

• tunnel chart -

- step-by-step process dikhata hai
- Jab stages ka conversion dikhana ho
- stage-wise values.



• waterfall chart -

- value ke positive/negative change dikhata hai
- Jab start to end value ke change explain karna ho e.g. stock market graph
- categories, values.



• scatter chart -

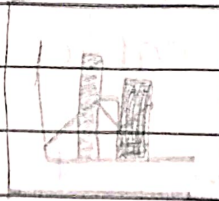
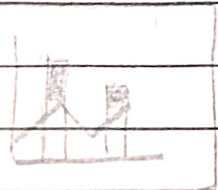
- do numerical values ka relation dikhata hai
- Jab correlation ya pattern dikhana ho
- x value, y value, optimal: size/c.



• combo chart -

- Bar + Line chart ek saath
- Jab comparison karna ho 2 measures ka
- category, 2 numerical value.

line and
stacked
column
chart



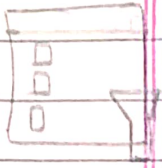
line and clustered
column chart

• Ribbon chart -

- show which data category has the highest rank. (largest value)



• Slicers - (It make interactive dashboard)



- filters provide karta hai report ke liye
- Jab user input chahiye filtering ke liye
- any categorical/ date field



(Slicer button)

• Q & A



- natural language se chart generate karta hai
- Jab user ko quick ad-hoc analysis karni ho
- all types (as per question)

*

Maps -

•

map -



- location - based data dikhata hai
- Jab states, countries ka data ho
- location + values

•

filled map -



- geographical regions ko fill karta hai colors se
- same as map, par filled regions ke liye
- location + value

*

Decomposition tree -



- measure ko drill down karke explain karta hai
- Jab breakdown analysis karna ho
- all types (as per question)

* Dashboard planning -

- requirement
- data overview
- data heading

① Title

② tiles

③ canvas size

* Canvas settings -

Go to visualization other format

Canvas

- canvas size

- canvas background

- wallpaper

* Dashboard creation -

- Go to insert

* Data modelling Overview -

Table and its types -

fact table -

- all transactions or measurable data usually numbers are presence here and it used visuals e.g. sales, quantity
- every row represent event or transaction
- It link table dimension to use foreign key

Ex., sales table: salesID, customerID, productID.

foreign key - It is set of attribute in a table that refers to the primary key of another table

2) dimension table -

- It is descriptive or lookup table it helps to explain fact table of rows.
- They have usually text-based information.
- They create filter context for power BI visuals.

e.g. customer Table: customerID, customerName, city, state.

3) Bridge Table -

- when two dimension tables between many-to-many relationship then a bridge table created.

e.g. studentcourse bridge: studentID, courseID

4) Date table -

- It important for time intelligence function in power BI.

- It have custom control.

e.g. date, month, year, quarter.

5) Lookup table (extended dimensions) -

- inner breakdown in dimension table when it used.

- It define hierarchy

e.g. category table: categoryID, categoryName
linked to: Product Table (via category ID)

6) Helper / utility table -

- calculated / manually created table used to special filtering in visuals also sorting and display.

Primary key - It is a column in a relational database table that is distinctive for each record (diff)

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- It don't have actual data only have control logic.

e.g. KPI level table: score range, performance band (good, average, poor)

§) summary table -

Type	Purpose	Example
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• Positioning and its type -

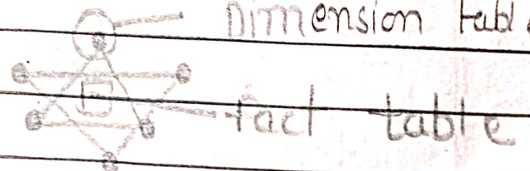
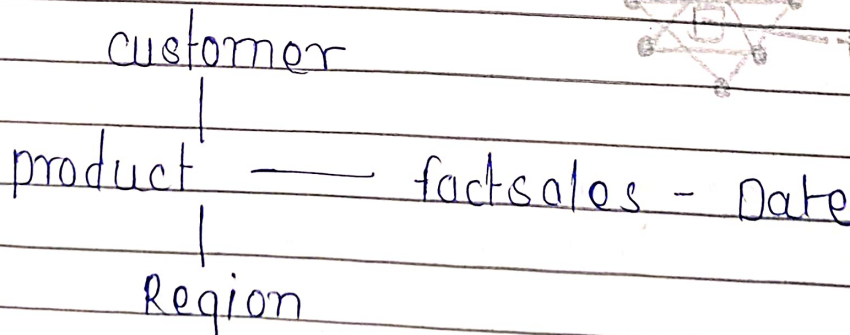
- positioning in power BI is most important especially when data model is large.
- it helps understand the data model, maintain and ignore the DAX errors.

1) star schema -

EMP

- It is preferred and optimized layout in power BI.
- There is one central fact table in that multiple dimension tables surround.
- It is simple and clean model with fast performance.

e.g.



TMP

2) snowflake schema positioning

- It is extended version of star schema
- In this schema, dimension table link with dimension table.
- most of normalization which means data redundancy are low but complexity are high
- performance show to compare star schema
e.g. product table → category table → Department table

3) flat table (single table model)

- when all data in one table without any relationship.
- useful for small report
- flexibility low when write the DAX
e.g. sales^{table} With: SalesID, Date, CustomerName, productname.

• Best practice for table positioning

- fact table should be at the center in your

5) Visual Hierarchy sug model view

- dimension table should be placed on the top and sides
- avoid creating relationship
- always keep one-directional flow
- maintain a clean layout
- similar tables take one side
- keep bridge table between two related table

• Visual Hierarchy suggestion

- Top level (Dimension table)
- middle level (bridge table)
- Bottom level (fact table)
- side/helper (utility table)

Keys and its types -

- with the help of keys we make relationship between tables.
- keys basically one or more columns which connect tables basis on it.

1) Primary key -

- unique column in table
- each row have diff value.
- eg. customerID.

2) foreign key -

- They match with other table primary key.
- eg. order table main customerID.

• use key in relationship -

- one table have primary key.
- other table have foreign key (also present repeated values)
- then power BI automatically ^{or} manually one-to-many or many-to-one create relationship.

• how to understand which key are -

- unique values → primary key.
- repeat values and match with other table → foreign key.
- In power BI 'manage relationship' option show match.

• Importance of key -

- accurate reporting
- save to wrong connection or duplicated data.

flow of relationship creation in power BI

- step 1: Tables load
- step 2: understand table structure
- step 3: Identify primary or foreign key
- step 4: go to 'model' view
- step 5: drag or drop then connect keys
- step 6: check relationship direction or cardinality (one-to-many)
- step 7: Create data analysis visuals

Relationship and its types -

- In power BI, tables create make model correctly because of them we create relationship between them.

- relationship link the data then we generate accurate visuals or insights.
- every relationship have proper flow, it uses cardinality, direction, keys.

1) one-to-many (1:*) -

- most common
- one table have unique values (primary key) another table have repeat values (foreign key)

2) Many-to-one (*:1) -

- opposite flow of one-to-many
- use to visualization or filter direction

3) one-to-one (1:1) -

Relationship backbone of Power BI
It helps to create accurate report-

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- Both table have unique column.

4) Many-to-many (*:*) -

- Both table have duplicate values or direct match.
- use carefully.

- Important tips while handling relationship -

- every relationship have one column unique (primary key)
- avoid circular reference
- carefully handle many-to-many relationship use intermediate bridge when they needed.
- active relationship occur one time when between one pair.
- when relationship properly defined RELATED, RELATEDTABLE are work.

Practical -

In relationship, dark line show most accurate relation

- DAX Overview -

- Data Analysis Expression (full of DAX)
- It is formula language used in Power BI, Power Pivot and SSAS (SQL server analysis services)
- calculated columns, measures and with queries data analysis this is main purpose of DAX.

- DAX formulas-

This is expression we write in calculated columns and measure

- DAX Queries-

- It is advance level of DAX statement returns the tabular data for eg. SQL query
- most use of DAX studio and advanced modeling.

- parameters-

parameters use for inputs and dynamic filtering.

It is not directly part of DAX but useful for modify queries.

- Measures-

It is fields of calculated to get aggregation (SUM, AVERAGE, COUNT) when calculated refresh the report then calculate it and use in visuals.

- DAX formulas Vs Excel formulas-

- Similarity

Both based on formulas function.

similar syntax (IF, sum, average, etc)

- Difference

- excel individually calculated every row but DAX filter context base on aggregate get.
- context important in DAX (row context, filter context)

Aggregation - a collection, or the gathering of things together

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- DAX work on relational model using relationship.

• Use of DAX

data analysis beyond basic visuals -

- custom calculations (Year to Date, Running total, etc).
- Business KPIs
- Advanced logic (eg, Top N, RANKX, CALCULATE)

• measures and calculated columns

- measures

dynamically calculated with visuals.
use for aggregation.
fast and optimized.

- calculated columns -

calculated on data load time.

row-wise calculation

have many memory.

• DAX formulas workflow -

* Excel Approach -

- formula in excel:

= SUM(Sales[Amount])

This formula gives total of visuals rows.

- when we applied slicer and filter ~~to~~ then they filter rows and formula.

return the sum of filtered rows.

- only work on visible rows. They can not understand context of slicer.

DAX are very flexible

* DAX approach-

- similar syntax formula

Total Sales = SUM (Sales [Amount])

- but use filter context concept.

- all table show (not filtered like excel)

- automatically applied DAX formula.

Aspect	Excel	DAX (power BI)
formula	= SUM (Sales [amount])	Total sales = SUM (Sales [Amount])
Before filter	1,00,000 (all regions)	1,00,000 (all regions)
After filter	20,000 (visible row only)	20,000 (filtered context)
formula reaction	same formula, unaware of filter	same formula result changes with filter.
view	filtered row only	full table, filtered result

Expression

सूत्र

विधि

formula

मार्गदर्शक

साधन

eg. DAX

• Aggregation formulas-

- Go to table tool

- new measure

- enter

- go to report view

- see data pane added new measure
eg Average sale = AVERAGE (UnitPrice)

count = Number, DATE
 countA = text, character
 count blank = blank

Distinct (distinct)

IMP
 • logical DAX -

Evaluate condition and return results based on whether those condition are true or false. These function are fundamental for creating conditional logic within measure and calculated columns.

* If formula -

= if (logical test, value if true, value if false)

Example -

- Add new column
- take new measure (table tools)
- Sale status = IF (sheet1[Totalsales] >= 1000, "High", "Low")

* And formula -

Example -

- ADD combination = AND(sheet1[quantity] >= 4, sheet1[unitPrice] >= 200)

* OR formula -

Example -

- OR combination = OR(sheet1[quantity] >= 4, sheet1[unitprice] >= 200)

* Iseven formula -

Example -

- iseven = ISEVEN (sheet1[quantity])

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 I I

- to Text DAX -

- Also we do in transformation
- new column add કરું જર ત્યામહે દાખવાયન
અમેલ તર new measure મહે જાકન કરાયન
જાહીતર transformation પણ આપન કરુ શકતો
- e.g., column = LEFT(sheet1[Region], 3)

- concatenate formula -
(જોડણી)

e.g., Column 2 = CONCATENATE (sheet1
[category], sheet1 [Product])

- Time and Date formulas -

- Date -

Edate
(end
date) dateview = TODAY()

- Time -

Newtime = NOW()